

Mechanisms and pathways of lateral flow on aspen-forested, Luvisolic soils, Western Boreal Plains, Alberta, Canada

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Abstract:

Rainfall simulation experiments by Redding and Devito (2008, *Hydrological Processes* **23**: 4287–4300) on two adjacent plots of contrasting antecedent soil moisture storage on an aspen-forested hillslope on the Boreal Plain showed that lateral flow generation occurred only once large soil storage capacity was saturated combined with a minimum event precipitation of 15–20 mm. This paper extends the results of Redding and Devito (2008, *Hydrological Processes* **23**: 4287–4300) with detailed analysis of pore pressure, soil moisture and tracer data from the rainfall simulation experiments, which is used to identify lateral flow generation mechanisms and flow pathways. Lateral flow was not generated until soils were wet into the fine textured C horizon. Lateral flow occurred dominantly through the clay-rich Bt horizon by way of root channels. Lateral flow during the largest event was dominated by event water, and precipitation intensity was critical in lateral flow generation. Lateral flow was initiated as preferential flow near the soil surface into root channels, followed by development of a perched water table at depth, which also interacted with preferential flow pathways to move water laterally by the transmissivity feedback mechanism. The results indicate that lateral flow generated by rainfall on these hillslopes is uncommon because of the generally high available soil moisture storage capacity and the low probability of rainfall events of sufficient magnitude and intensity. Copyright © 2010 John Wiley & Sons, Ltd.

KEY WORDS Boreal Plain; subsurface flow; preferential flow; antecedent soil moisture storage; transmissivity feedback

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INTRODUCTION

Subsurface flow from hillslopes is typically initiated when precipitation infiltrates the soil and moves vertically to an underlying flow restricting or impeding layer where it is redirected as lateral flow above the layer of restricted permeability (McDonnell, 1990; Montgomery *et al.*, 1997; Weiler *et al.*, 2005). In many areas of the world where subsurface runoff mechanisms have been studied, lateral flow occurs at the soil–bedrock interface (e.g. McDonnell, 1990; Peters *et al.*, 1995; Tromp-van Meerveld and McDonnell, 2006). In areas without bedrock or another obvious restricting layer, such as many glacial landscapes, the mechanism of lateral flow generation may not be obvious (Lischeid, 2008). It is often proposed that where soils feature a coarser-textured horizon over a finer-textured horizon, lateral flow will occur at that interface (Lin *et al.*, 2005).

The soils found on aspen-forested uplands on the Boreal Plain region of Alberta feature a relatively coarse-textured Ae horizon (clays and organic matter eluviated) overlying a clay-enriched Bt horizon (Whitson *et al.*, 2004) and are classified as Luvisolic (Soil Classification Working Group, 1998). In other parts of the world where soils feature a coarse-textured A horizon overlying

a finer-texture B horizon, lateral flow at the A–B interface has not always been observed (Wilson *et al.*, 1990; Newman *et al.*, 1998; Lin *et al.* 2005; Lohse and Dietrich, 2005). Whitson (2003) and Whitson *et al.* (2004) studied lateral flow on aspen-forested hillslopes with Luvisolic soils on the Boreal Plains of Alberta. They found that lateral flow at the Ae–Bt interface was not significant (<1 mm) during the snowmelt period or summer rain-storm events, but their measurements were made during years with below average precipitation. However, Whitson *et al.* (2005) speculated that lateral flow was occurring at the Ae–Bt horizon interface, on the basis of their interpretation of vertical tracer profiles. A review of the literature on hillslope runoff on Luvisolic soils identified only a single instance where significant lateral flow was measured during snowmelt at the Ae–Bt interface; this was in Minnesota (Timmons *et al.*, 1977). However, it was unclear whether soil frost contributed to the development of transient perching which occurred during snowmelt. In addition, the contribution of lateral flow at the Ae–Bt interface to the annual water budget was only a small proportion (approximately 7%) of annual catchment outflow. On the basis of groundwater data for an aspen-forested Boreal Plains watershed during a large storm, Devito *et al.* (2005a) hypothesized that rapid vertical preferential flow to the clay-rich C horizon resulted in the development of a perched water table and that lateral flow was generated via the transmissivity feedback mechanism (Rodhe, 1987). Transmissivity

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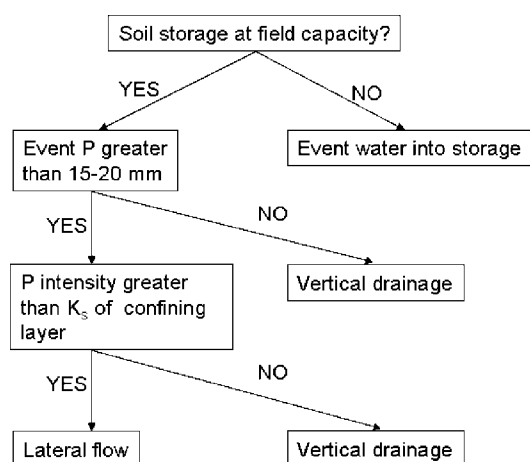


Figure 1. Threshold-based conceptual model of lateral flow generation

feedback occurs when the water table rises upward in the soil profile to intersect layers of higher transmissivity, resulting in greater lateral flow which simultaneously keeps the water table from rising higher (Rodhe, 1987).

Redding and Devito (2008) used rainfall simulation experiments to examine lateral flow thresholds for an aspen-forested hillslope with Luvisolic soils and noted that lateral flow occurred mainly as preferential flow through the Bt horizon rather than at the Ae–Bt interface. This paper builds upon the results of Redding and Devito (2008) who found that on forested hillslopes with Luvisolic soils on the Boreal Plains, lateral flow generation is a function of P intensity relative to soil K_s and antecedent soil moisture storage. The conditions required for event-scale lateral flow generation under rainfall conditions are summarized in Figure 1. This paper presents detailed analysis of pore pressure, soil moisture and tracer data from the rainfall simulation experiments, which is used to identify lateral flow generation mechanisms and flow pathways. The experimental design included two adjacent plots with subtle differences in soil properties and contrasting antecedent soil moisture conditions. The specific questions addressed herein are:

1. What is the lateral flow generation mechanism and flowpath?
2. What is the potential for lateral flow from these hillslopes under natural conditions?

The experimental approach allows for testing of specific hypotheses about lateral flow generation mechanisms and flowpaths, and it provides an indication of when lateral flow occurs on these and similar hillslopes (Kirchner, 2006; Lischeid, 2008). This paper and Redding and Devito (2008) do not consider the potential for snowmelt generated lateral flow, which may demonstrate different controls and thresholds (Redding, 2009). Understanding the mechanisms, flowpaths and timing of lateral flow generation on forested hillslopes is critical for predicting the effects of anthropogenic disturbances, such as forest harvesting and oil and gas development, on hydrological processes and biogeochemical exports to receiving

wetlands and surface waters. For example, the research described herein may be used as an undisturbed analogue against which to evaluate mining reclamation practices in the Alberta oil sands region with similar soils and climate (e.g. Elshorbagy *et al.*, 2005).

STUDY SITE

The rainfall simulation experiments were carried out at the Utikuma Region Study Area (URSA) near Utikuma Lake (lat: 56°N, long: 115°30'W) in north-central Alberta (Devito *et al.*, 2005b). The site is located within the Boreal Plains ecozone. Mean annual precipitation and potential evapotranspiration are 481 and 518 mm, respectively (Marshall *et al.*, 1999). The climate at the study site is classified as subhumid in the Canadian Ecoregion system (Ecoregions Working Group, 1989).

The rainfall simulation plots were established on a hillslope with Grey Luvisolic soils (Soil Classification Working Group, 1998) that developed from disintegration moraine deposits. These soils are typically silt rich but spatially heterogeneous with zones of high clay or sand content (Fenton *et al.*, 2003). The overstory canopy of the study hillslope is predominantly trembling aspen (*Populus tremuloides*) with minor amounts of balsam poplar (*Populus balsamifera*) and willow (*Salix* spp.) in depressions and draws. The understory vegetation at the site includes prickly rose (*Rosa acicularis*) and high-bush cranberry (*Viburnum edule*) in the shrub layer, and twinflower (*Linnaea borealis*) in the herb layer. The forest floor surface is dominated by aspen litter with small amounts of moss and lichen. A more detailed description of the study site is available in Redding and Devito (2008).

METHODS

Detailed descriptions of the methods used to establish the experimental plots, conduct the rainfall simulations, characterize soils and measure lateral flow and soil moisture are provided in Redding and Devito (2008). Briefly, a 1.55 × 1.9 m plot was trenched and excavated to 0.75 m depth in mid-July 2004 [DOY (day of year) 204]. The perimeter of the plot was trenched and sealed with plastic, excluding roots from trees and larger shrubs and resulted in increased soil moisture prior to the experiment. This plot was termed the root exclusion (RE) plot. A second plot of 1.7 × 2.0 m plot was excavated to 0.75 m on 16 September 2004 (DOY 260), allowing root uptake (RU plot) during the 2004 growing season prior to the start of the rainfall simulations (DOY 261).

Within each plot, soil pore pressure was measured with tensiometers (Soil Measurement Systems, Tucson, AZ, USA) connected to data loggers (CR10X, Campbell Scientific Inc., Logan, UT, USA). The tensiometers were installed approximately 1.25 m upslope from the downslope face of the soil columns, and at depths of 0.3 and 0.5 m in the RE plot and 0.1, 0.3, 0.5 and 0.75 m in

the RU plot. The tensiometers were installed in vertical boreholes that were the same diameter as the tensiometer body. The porous cup was seated in a slurry of native soil from the appropriate depth and was sealed with bentonite at the soil surface. Above-ground sections of the tensiometers were insulated to minimize temperature effects. Soil moisture and pore pressure measurements were recorded every 10 min during the 8 day irrigation experiment period and every 2 h during the pre- and post-irrigation monitoring periods. All TDR probes and pressure transducers were individually calibrated prior to the experiment. Unsaturated hydraulic conductivity was calculated from tension infiltrometer data using the TIDAP-6 software (Reynolds, 1995). Macroporosity (%) was estimated following Buttle and McDonald (2000).

Subsurface flow (matrix and macropore flow) was collected in troughs inserted at the horizon boundaries [FF–Ae, Ae–Bt, Bt–BC (C)] and at the base of the pit in the C horizon (Redding and Devito, 2008). To quantify subsurface matrix and macropore flowpaths, Cl^- and Br^- were added to irrigation water as tracers. The Cl^- tracer was added as KCl to the first irrigation event (30 l) at a concentration of 4100 mg l^{-1} ($[\text{Cl}^-] = 985 \text{ mg l}^{-1}$). The Br^- tracer was added as LiBr to the seventh irrigation event (120 l) at a concentration of 2000 mg l^{-1} ($[\text{Br}^-] = 920 \text{ mg l}^{-1}$). Lateral flow was expected to occur during this event. The Br^- tracer was used to quantify the pre-irrigation and irrigation water fractions in lateral flow. For each water sample collected, the proportion of water from the irrigation events was calculated on the basis of the change from the initial concentration of Br^- in the irrigation water. More complex mixing models were not applicable as Br^- in the soil was below the analytical detection limit, and between-event concentrations of soil solution Cl^- were not measured. To calculate the overall event water proportion, the Br^- mass recovered in the lateral flow was compared with the total amount applied.

Subsurface flow velocities were estimated on the basis of the time interval between the start of the irrigation application and the collection of the first lateral flow sample for the shortest straight-line distance between the soil surface and the lateral flow collection trough of interest (the vertical distance from the forest floor surface to the trough). The minimum vertical distance was used to calculate the velocity because it was not possible to determine the surface origin of the water that emerged at the pit face. Therefore, the calculated value is a conservative estimate of velocity because the shortest flowpath length was used and the emergence of preferential flow was noted visually on the pit face prior to collection of the first sample.

RESULTS

Soil hydraulic properties

There was a general decrease in saturated hydraulic conductivity (K_s) with depth on the study hillslopes (Figure 2). In the hillslope soils adjacent to the two runoff

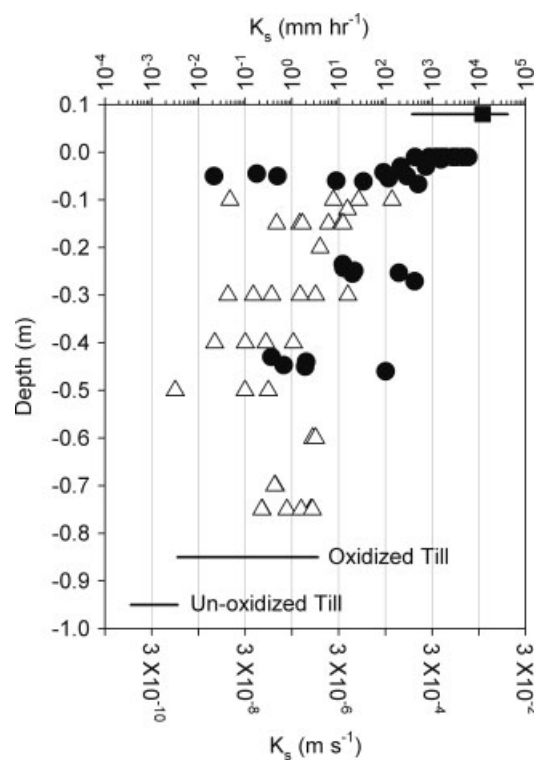


Figure 2. Field measured saturated hydraulic conductivity (K_s), determined *in situ* using the single ring infiltrometer (filled circle) and Guelph permeameter (open triangle) methods. Values (mean and range) for forest floor (filled square) are taken from Lauren and Heiskanen (1997). The range in K_s for oxidized and unoxidized till materials is taken from Hendry (1982), Hayashi *et al.* (1998), and Ferone and Devito (2004), as measured at other Boreal Plains and Prairie locations with similar surficial geology

plots, K_s was highly spatially variable but variability decreased with depth (Figure 2). The K_s below 30 cm depth ranged from 10^1 to $10^{-3} \text{ mm h}^{-1}$, and this is similar to the range observed in the oxidized layer which extends from 3 to 7 m or more at sites with similar geology (Hendry, 1982; Hayashi *et al.*, 1998; Ferone and Devito, 2004). Higher rates of K_s were measured at the same depths with the single ring method than with the Guelph permeameter, indicating the contribution of macropores (e.g. Buttle and House, 1997). Across the study hillslope, measured macroporosity was higher at the forest floor (FF) surface than at the underlying mineral soil surface. The calculated mean macroporosity from the tension infiltrometer measurements at the FF and mineral soil surfaces were 0.01% and 0.006%, respectively.

Soil moisture conditions necessary for subsurface flow

To generate lateral flow, it was necessary to first satisfy the storage deficit required to bring the soils roughly to field capacity and then apply a threshold amount of precipitation—in this case, 15–20 mm (Redding and Devito, 2008). Whereas the RE plot soils were wet prior to irrigation, available storage above the C horizon needed to be satisfied in the RU plot before lateral flow was generated (Figures 3 and 4). In the RE plot, where tree root uptake was eliminated, wetting occurred during the pre-irrigation period (Redding and Devito, 2008);

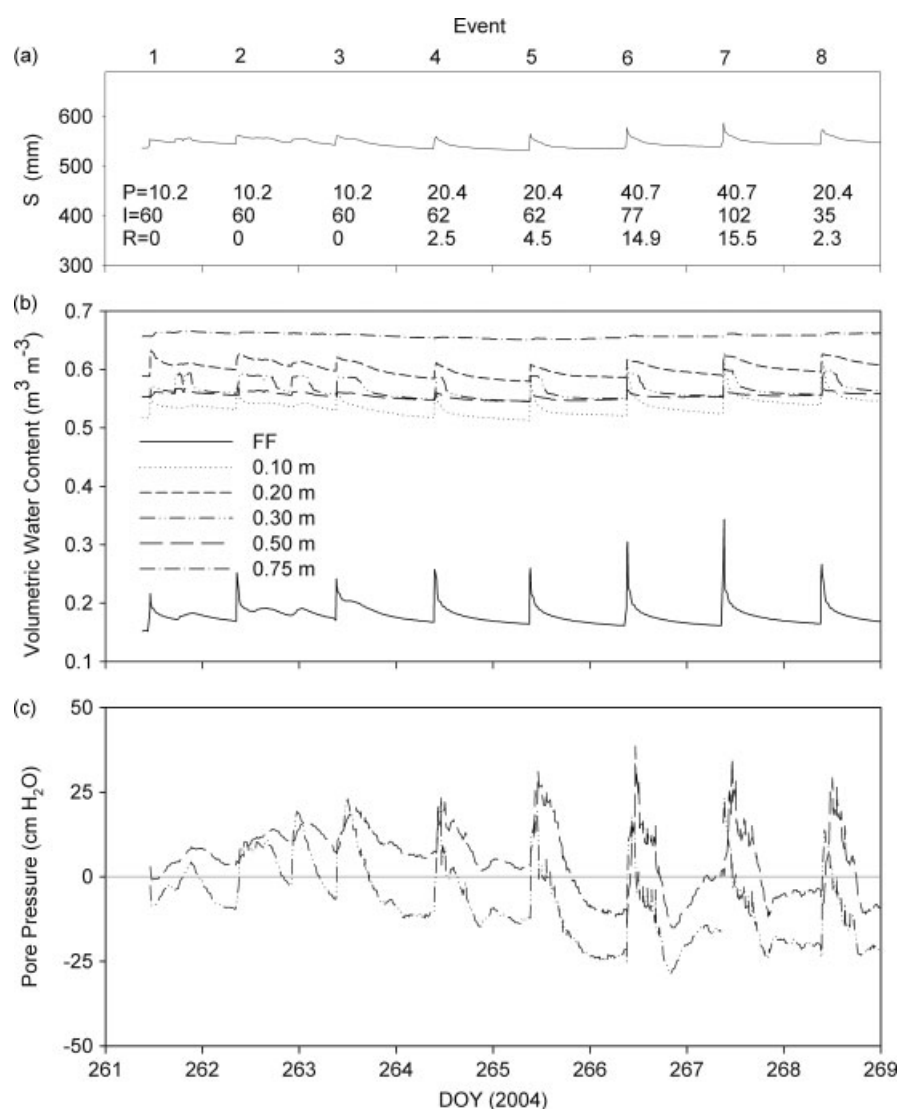


Figure 3. Root exclusion (RE) plot: (a) soil moisture storage (S), (b) volumetric water content, and (c) pore pressure over the duration of the experiment (DOY: day of year). Pore pressure data immediately prior to event 7 are missing due to the refilling of the tensiometers. Numbers along the top of (a) refer to event precipitation (P , mm), irrigation intensity (I , mm h⁻¹), and runoff (R , mm) for each event. In panel (b), FF refers to forest floor. Panel (b) is modified from Redding and Devito (2008)

however, no lateral flow was generated during this period. The RE plot generated lateral flow for all events ≥ 20 and 35 mm h⁻¹ in intensity (Figure 3). At the start of the irrigation experiments, the soil in the RU plot was wetted to field capacity to a depth of 0.3–0.5 m; however, lateral flow was not generated until soils were wetted to the C horizon (0.5–0.75 m) during irrigation event 5 (cumulative precipitation $P = 62$ mm) (Figure 4). The wetting front reached 0.5 m during event 4 and 0.75 m in event 6 (Figure 4).

Lateral flow mechanisms

Wetting of the soil profile occurred from the top down during all irrigation events. Event 7 produced the largest lateral flow response in both plots; therefore, it is used here as an example of the interactions of lateral flow, soil moisture and pore pressure response. Subsurface flow in the RE plot occurred predominantly through the Bt horizon (B trough, 11.2 mm), with smaller contributions

from the BC horizon (C trough, 4.0 mm) and then the Ae horizon (A trough, 0.4 mm) (Figure 5). Peak lateral flow rates occurred earlier for the A and B troughs than the C trough, in contrast to the RU plot where all three peaked simultaneously (Figure 6). For the RE plot, lateral flow into the B and C troughs persisted for almost 200 min from the time irrigation began. As with the RU plot, flow from the RE plot A trough occurred only during one measurement interval, whereas flow from the C horizon (C trough) ended at the same time as the B trough (Figure 5). Tensiometers responded first at 0.3 m, then at 0.5 m (Figure 5). The maximum pore pressure for the 0.3 m tensiometer (+75 cm H₂O) occurred prior to the peak in lateral flow rate and TDR response at any depth (Figure 5), possibly indicating a pressure wave-type response. Saturation at the 0.3 m depth persisted for approximately 160 min after the start of the irrigation application. At the 0.5 m depth, the soil moisture and pore pressure responses occurred

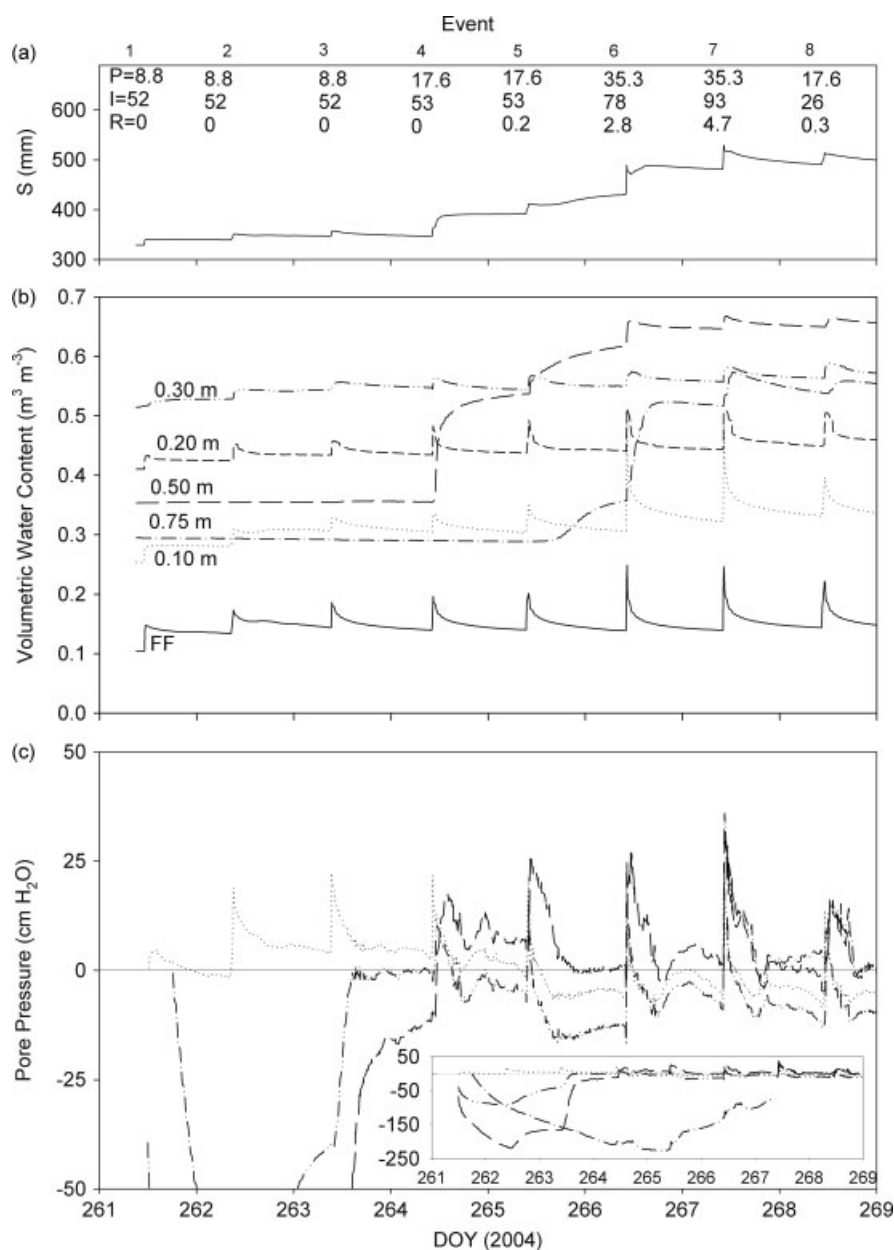


Figure 4. . Root uptake (RU) plot: (a) soil moisture storage (S), (b) volumetric water content, and (c) pore pressure over the duration of the experiment (DOY: day of year). Pore pressure data prior to event 7 are missing due to the refilling of the tensiometers. Numbers along the top of (a) pertain to event precipitation (P , mm), irrigation intensity (I , mm h⁻¹), and runoff (R , mm) for each event. The line types for each measurement depth are consistent between panels (b) and (c). The inset in panel (c) shows the full range of pore pressures as the deeper tensiometers equilibrated. In panel (b), FF refers to forest floor. Panel (b) is modified from Redding and Devito (2008)

at approximately 20 min after the start of irrigation (Figure 5). The response was muted because the soils were already very close to saturation. Approximately 50 min after the start of irrigation, the 0.5 m tensiometer showed an increase in pore pressure with no concurrent increase in soil moisture content. This pore pressure response is likely related to the increase in air temperature during the morning (e.g. Warrick *et al.*, 1998).

In the RU plot during event 7, lateral flow occurred predominantly through the Bt horizon (B trough, 3.0 mm), with smaller contributions from the C horizon (C trough, 0.9 mm) and Ae horizon (A trough, 0.8 mm) (Figure 6). Lateral flow into the B trough persisted for almost 100 min from the time irrigation began. Flow

from the A trough occurred only during one measurement interval, whereas flow from the C horizon (C trough) ended at the same time as the B trough (Figure 6). After the start of irrigation, tensiometers responded in the following depth order: 0.1, 0.3, 0.5 and 0.75 m (Figure 6). Tensiometer and TDR responses peaked at the 0.1 and 0.3 m depths at the same time as the peak lateral flow rate, whereas a slight lag occurred at 0.5 and 0.75 m. All depths became saturated (pore pressures >0 cm H₂O). Saturation at the 0.10 and 0.30 m depths persisted for 120–150 min following the start of irrigation, whereas the response of the deeper tensiometers persisted for more than 250 min (Figure 6), indicating the presence of saturated conditions above and within the upper C horizon.

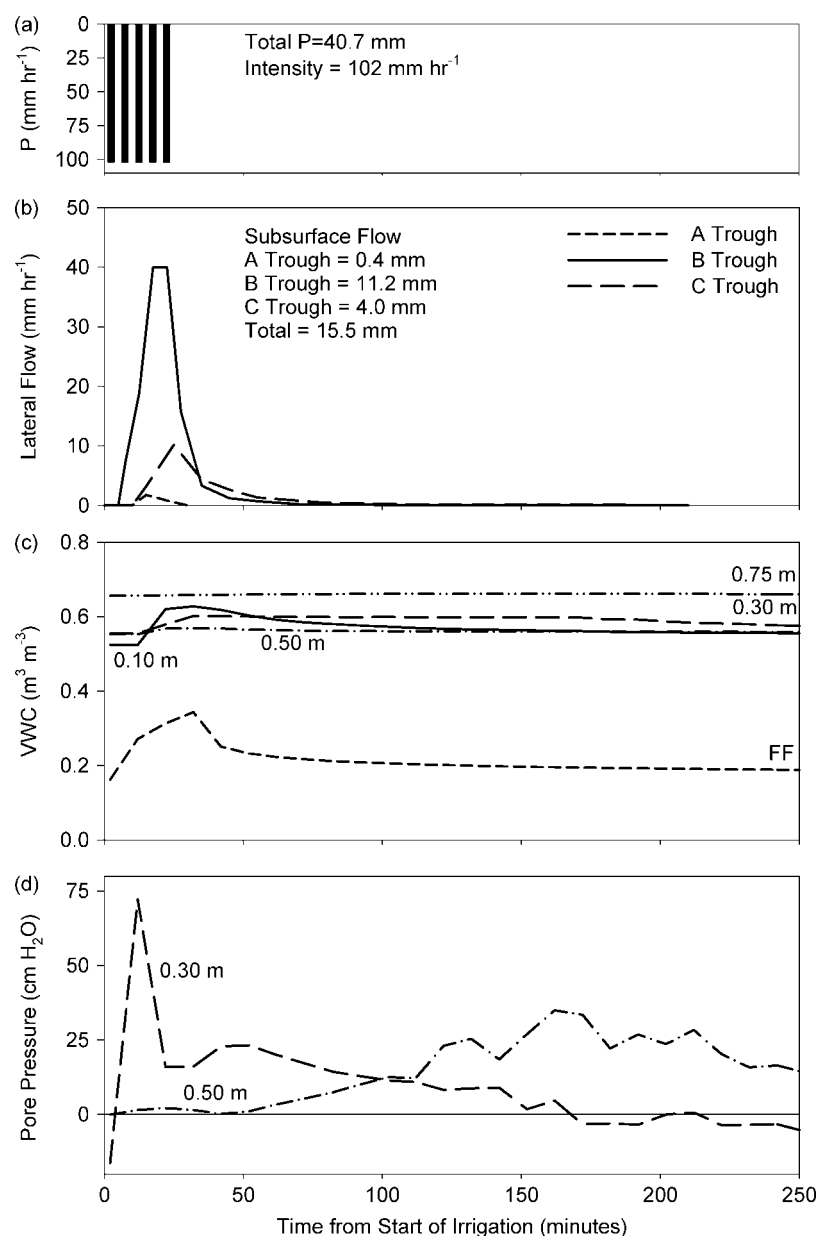


Figure 5. Root exclusion (RE) plot: event 7 time series for (a) precipitation intensity, (b) subsurface flow rates, (c) volumetric water content (VWC), and (d) pore pressure. The line types for each measurement depth are consistent between panels (b) and (c). For the sake of clarity, the volumetric water content data at 0.2 m depth are not shown

While detailed results are presented only for event 7, the pattern of wetting from the surface downward was similar for all events (Figures 3 and 4).

In the RU plot during events 3 through 5, soil moisture content at 0.75 m depth, measured by TDR, was constant; however, there was a transient pressure response from the tensiometer at 0.75 m at the time of each irrigation application (Figure 4), although it was relatively short-lived. The magnitude of the pressure response increased between event 3 and 5 (from about +5 to +25 cm H₂O) as the wetting front moved deeper.

Flow from the pit face began rapidly during events in which lateral flow occurred. Lateral flow from the pit face was initiated as preferential (macropore) flow, primarily along root channels, and then the pit face appeared to wet up outward from the macropore locations

(T. Redding, personal observation). Within the RE plot, nine preferential flow outlets were noted on the pit face; all were associated with tree roots (Figure 7), and the order of flow initiation and cessation (reverse of pattern of initiation) was consistent among events. Within the RU plot, six preferential flow outlets were noted, with none in the C horizon (Figure 7). Similar to the RE plot, the pattern of initiation and cessation was consistent among events. Although no preferential flow from the C horizon was noted in the RU plot, sand lenses (3–10 mm wide), vertical cracks (<2 mm wide) and fine roots (1–2 mm wide) were visible in the pit face (T. Redding, personal observation).

Event water dominated the subsurface flow response to event 7 in both the RE and RU plots (Table I). In both the RE and RU plots, peak Br⁻ concentration corresponded

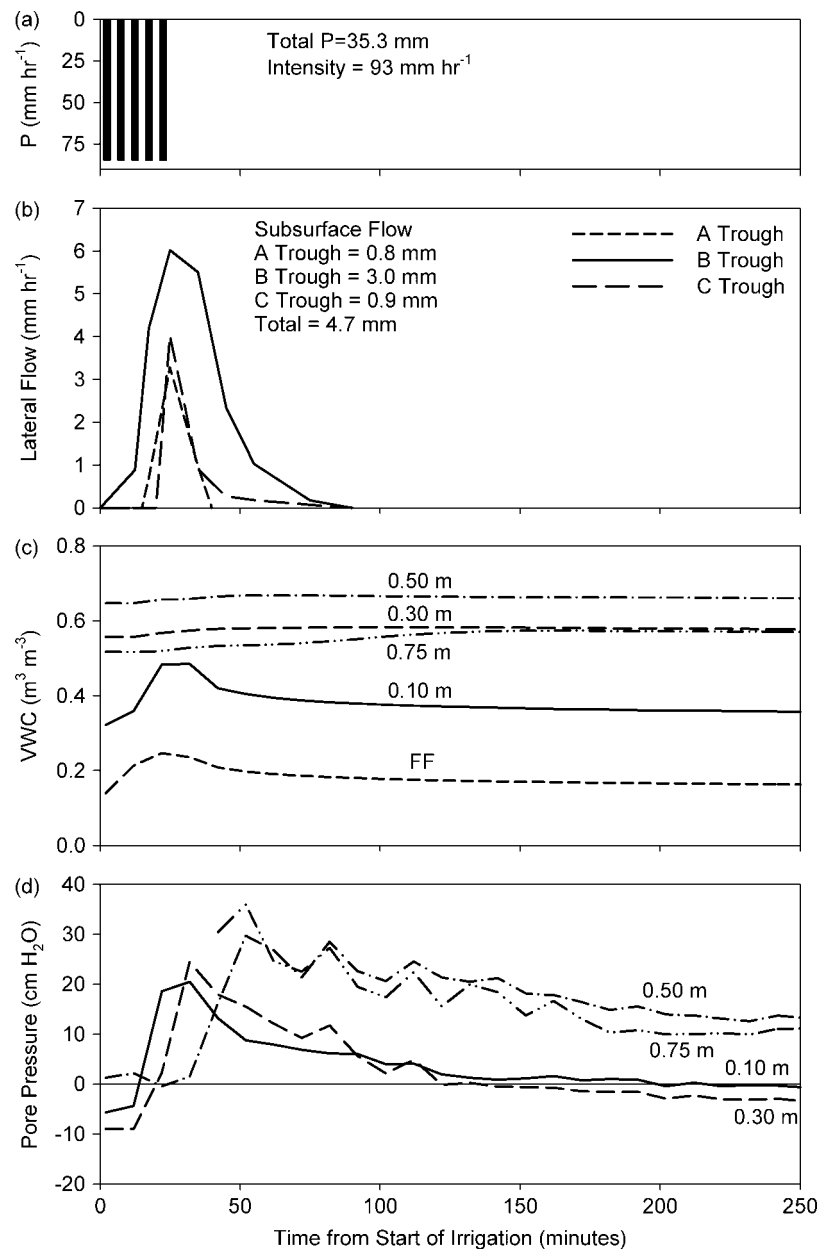


Figure 6. Root uptake (RU) plot: event 7 time series for (a) precipitation intensity, (b) subsurface flow rates, (c) volumetric water content (VWC), and (d) pore pressure. For the sake of clarity, the volumetric water content data at 0.2 m depth are not shown. Note the different scale of subsurface flow compared to Figure 6

with peak outflow rates (Figure 8). As the irrigation event progressed, the role of pre-event water became more pronounced, as indicated by a decrease in Br^- , which was added in event 7, and by an increase in Cl^- , which was present in the soil matrix. The maximum event water contributions were approximately 75% early in the event and decreased to about 25% at the cessation of subsurface flow (Figure 8). The total event water contribution for event 7 was 61% and 60% for the RE and RU plots, respectively, which corresponded to 9.5 and 2.8 mm of subsurface flow, respectively (Table I). Event water outputs were higher for the A and B troughs than for the C trough (Table I), possibly reflecting differences in preferential flow contributions to outflow. On the RU plot, the C trough event water contribution was 52% of

the flow from this horizon (0.5 mm). On the RE plot, preferential flow emerged from the C horizon during events 6 and 7 by way of root channels. Flow from this horizon was 47% event water during event 7 (Table I).

Table I. Event water contributions (%) for event 7 in the root exclusion (RE) and root uptake (RU) plots

Trough	RE plot	RU plot
A	61% (0.2 mm)	62% (0.5 mm)
B	66% (7.4 mm)	60% (1.8 mm)
C	47% (1.9 mm)	52% (0.5 mm)
Total	61% (9.5 mm)	60% (2.8 mm)

Values are volume weighted event water % of lateral flow collected at each trough and total depth of event water in lateral flow (mm).

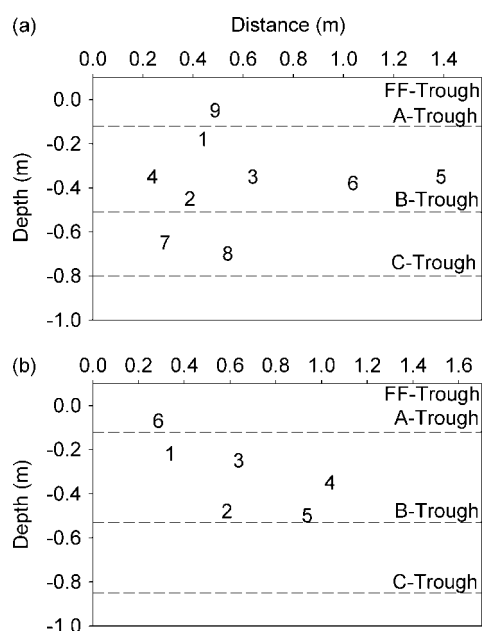


Figure 7. Locations of preferential flow outlets at the pit faces of the (a) root exclusion (RE) plot and (b) root uptake (RU) plot. Preferential flow outlets are numbered in the order in which they began contributing flow. FF refers to forest floor

Subsurface flow velocities were much greater than the soil K_s and ranged from less than 1000 mm h^{-1} for the A trough to a maximum of about 4000 mm h^{-1} for the B and C troughs in the RE plot (Figure 9). Subsurface flow velocities were positively related to irrigation intensity (Figure 9), and the relationship appears to be fairly linear over the range of applied precipitation intensities for the B and C horizons where the majority of the flow emerged. There were no systematic differences in subsurface flow velocity between the RE and RU plots for the range of applied precipitation intensities (Figure 9).

DISCUSSION

Lateral flow mechanisms and flowpaths

In the two experimental plots in this study, lateral flow was not generated along the Ae–Bt interface as has been hypothesized for Luvisolic soils (Wilson *et al.*, 1990; Lin *et al.*, 2005; Whitson *et al.*, 2005) or has been observed in Minnesota (Timmons *et al.*, 1977). The soils in this study have no clear flow restricting layer, and lateral flow was generated primarily by macropore flow through root channels in the clay-enriched Bt horizon, similar to other

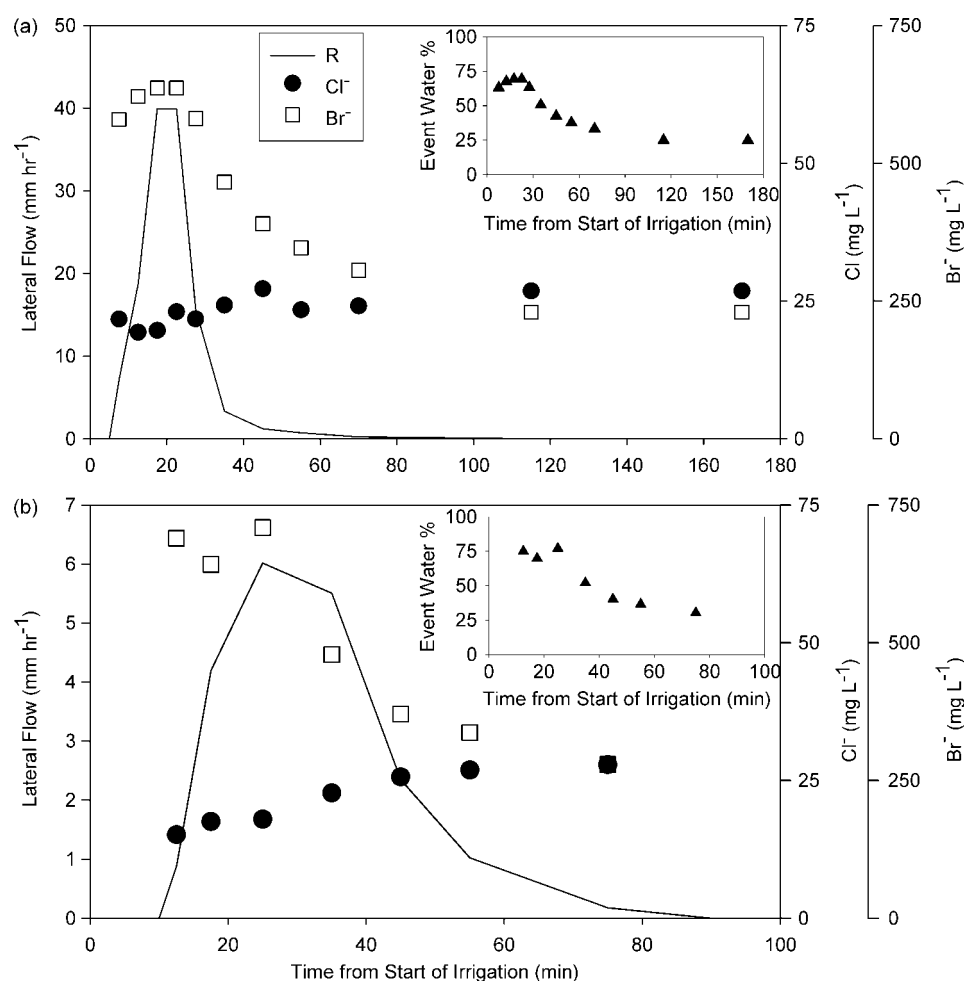


Figure 8. Event 7 (a) root exclusion plot (RE) B trough lateral flow rates and Cl^- and Br^- tracer concentrations, and (b) root uptake plot (RU) B trough lateral flow volumes and Cl^- and Br^- tracer concentrations. The inset plots show the time series of event water proportions (%) over the duration of flow. The Br^- tracer was applied during event 7 at a concentration of 920 mg l^{-1}

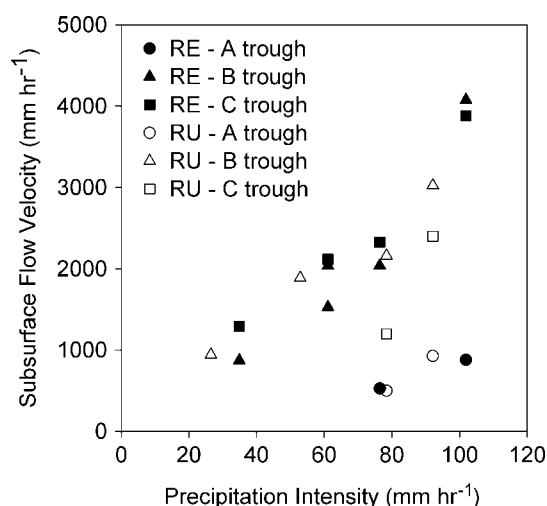


Figure 9. Calculated minimum subsurface flow velocities at each collection trough versus precipitation intensity for events with measured lateral flow. Filled symbols are for the root exclusion (RE) plot; open symbols are the root uptake (RU) plot. The A trough is denoted by circles, the B trough by triangles, and the C trough by squares

Boreal Plains catchments (Devito *et al.*, 2005a). The antecedent moisture conditions and precipitation intensities required to generate lateral flow in the experimental plots were much greater than those recorded by Whitson *et al.* (2004, 2005) during the lateral flow monitoring period on similar sites on the Boreal Plains. Similar to the results recorded in this study, Whitson *et al.* (2004, 2005) found no significant lateral flow (<1 mm) at the Ae–Bt interface during snowmelt or summer rainstorms, but they did not measure the potential for lateral flow below the Ae–Bt horizon interface.

Similar flow pathways to those measured in this study have been noted in semi-arid forests in the southwestern United States during periods with high antecedent soil moisture conditions. In New Mexico, Newman *et al.* (1998) found that 80% of lateral flow occurred through the Bt horizon, and lateral subsurface flow was infrequent from the hillslopes in the region. Similar to this study, lateral flow was dominated by preferential flow through root channels, and wetting of the unsaturated profile occurred outward from macropore channels rather than from saturation from below (Newman *et al.*, 2004). Peters *et al.* (1995) found that on the Boreal Shield, lateral flow occurred mainly through the B horizon and at the soil–bedrock interface.

If preferential flow in this study was initiated in the forest floor or near the mineral soil surface, the lateral preferential flow component may be a function of the shape of the rooting profile of the aspen trees. Roots tend to follow an inverse parabolic shape (umbrella like); thus, the pit face may have intersected the root channels where they were still trending laterally rather than deeper where they would have had a more vertical orientation deeper in the profile and not intersect the pit face. If this is the case, the plot measurements will overestimate lateral flow contributions at the hillslope scale. Root density decreases greatly with depth at the study site

(Redding and Devito, 2008), so there are also fewer roots intersecting the pit face below the Bt horizon. There is tracer evidence of deep, vertical preferential flow into the C horizon on both the RE and RU plots, which is likely associated with root channels (Redding and Devito, 2008) and or soil fractures (Hendry, 1982).

The large contribution of event water to lateral flow on both plots in this study differs from many other hillslope studies, which indicate that runoff response is dominated by water stored within the hillslope prior to a precipitation event (Buttle, 1994; Lischeid, 2008). The high event water contributions to lateral flow may be due to the initiation of preferential flow from near the surface early in the irrigation event, combined with low interaction between preferential flow water and the soil matrix early in the event (Kienzler and Naef, 2008). The hypothesis about limited interaction between preferential flow pathways and the soil matrix (e.g. Thomas and Phillips, 1979; Newman *et al.*, 1998; Weiler and Naef, 2003; Scherrer *et al.*, 2007) is supported in this study by the rapid transport of both water and tracers from irrigation inputs into lateral flow at the pit face. Wilson *et al.* (1990) noted that higher intensity precipitation produces a greater contribution of early event water through preferential flow and that the contribution of event water decreases through time. The increase in pre-event water during the progression of the event is likely due to the greater proportion of stored water contributing as the perched water table causes mixing. Kienzler and Naef (2008) reported similar findings of different event–pre-event water contributions for different flow pathways.

Preferential flow emerging from the Bt horizon in this study could have been caused by two different mechanisms, either working separately or in concert. The first mechanism is preferential flow initiation near the soil surface, followed by preferential flow both laterally and vertically through the soil profile. The second mechanism results from the development of a transient perched water table at the C horizon boundary due to conditions that result in percolation excess. The perched water table rises through the profile, causing the saturated zone to intersect with preferential flow pathways in the Bt horizon. This transmissivity feedback mechanism, which has been observed in many glacial till environments (e.g. Rodhe, 1987; Buttle, 1994; Kendall *et al.*, 1999), likely accounts for the increasing proportions of pre-event water as the event progresses (Figure 8).

Determining the specific location of preferential flow initiation in the upper soil profile is important for deciphering the potential proportioning of lateral and vertical flow pathways. In this study, preferential flow may have been initiated at the forest floor–mineral soil interface, given the larger macroporosity at the forest floor surface than at the mineral soil surface, and the shingling effect of overlapping aspen litter, which may concentrate flow and the high-intensity irrigation. In steep, coastal soils, De Vries and Chow (1978) found that preferential flow was initiated in the forest floor. Wilson

et al. (1990) hypothesized that in soils with profiles with similar Ae–Bt horizons, preferential flow is initiated at the Ae–Bt boundary due to transient perching resulting from the permeability contrast. Future studies using dye tracing methods may be able to clarify the location of preferential flow initiation (e.g. Weiler and Fluhler, 2004).

It is important to place the amount of lateral flow in the context of vertical flow and soil storage, which dominated the water movement in both plots in this study (Redding and Devito, 2008). At other sites on the Boreal Plains, the hydrological response to precipitation is dominated by vertical flow and storage and loss to evapotranspiration, with little annual recharge below the root zone (Cuenca *et al.*, 1997; Elliott *et al.*, 1998; Blanken *et al.*, 2001; Devito *et al.*, 2005a). Neither Elliott *et al.* (1998), Blanken *et al.* (2001) nor Cuenca *et al.* (1997) were able to account for vertical or lateral preferential flow in their measurements.

The subsurface flow velocities estimated in this study are within the range of those reviewed by Nimmo (2003) and measured by Tymchak and Torres (2007), Kienzler and Naef (2008), and Anderson (2008), but they are lower than those measured on forested New Zealand hillslopes with very large macropores (Mosley, 1982). The high velocities recorded in this study may indicate a pressure wave (Torres, 2002; Torres and Alexander, 2002) or kinematic wave response (Beven, 1982). Evidence of a transient pressure wave response was provided by the RU plot tensiometers where short-term pressure effects were noted at depth in unsaturated conditions. In contrast, TDR probes at the same depth did not respond in kind (data not shown).

The role of precipitation intensity

The results of the current study are similar to other studies in which precipitation intensity was important for initiating preferential flow near the soil surface (Weiler and Naef, 2003) and for partitioning lateral flow and deep percolation at the restricting layer boundary (Jardine *et al.*, 1990; Uchida *et al.*, 2005; Lehman *et al.*, 2007; Tromp-van Meerveld *et al.*, 2007). In this study, and in others on the Boreal Plains and Prairies (Hendry, 1982; Hayashi *et al.*, 1998; Devito *et al.*, 2005a), high forest floor and near-surface K_s exceeded both natural and simulated rainfall intensities, and together with large available soil storage, excluded lateral flow from the Ae–Bt horizon. Irrigation intensities during the latter events and observed subsurface flow velocities in the Ae and Bt horizons of the study plots were greater than the K_s of the C horizon. This allowed the C horizon to act as a restricting layer to downward infiltration, which lead to the accumulation of water in excess of deeper percolation, and the development of a transient perched water table. Water accumulating above the restricting layer could then back up within the profile and exit through the macropores, which were predominantly in the Bt horizon (Figure 7). Because the soil matrix was

effectively saturated (Figures 5 and 6), preferential flow moved water laterally from the profile.

It is possible that the high irrigation rates in this study influenced the runoff generation mechanisms (e.g. preferential flow initiated at the soil surface) and consequently enhanced the event water contributions to lateral flow relative to more common lower intensity natural events (Pearce *et al.*, 1986). Studies in New Zealand showed that artificial ponded infiltration (Mosley, 1979) resulted in much higher event water contributions to lateral flow than occurred under natural rainfall conditions (Sklash *et al.*, 1986). It is also important to note that the average intensity of the irrigation applications was lower than the instantaneous application rates due to the small application area of the watering cans. These high intensities may be important in determining where in the soil profile preferential flow is initiated and the event water contributions (Wilson *et al.*, 1990).

The amount and rate of lateral flow may have been affected by the high precipitation intensities. Uchida *et al.* (2005) found that once the threshold rainfall depth necessary for lateral flow generation was exceeded, the maximum rate of preferential flow was positively related to the precipitation intensity. Effects of high-intensity precipitation application on the soil moisture and pore pressure responses are likely small in the context of the current experiment. There were no changes in response as events became more intense, likely due to the effect of soils reaching field capacity, thus reducing the potential magnitude of response.

Given the large magnitude and high-intensity rainfall inputs necessary to generate lateral flow at these sites, it is likely that relatively large preferential flow contributions and event water proportions would be found under natural conditions. It is unknown how continuous the preferential flow network is within these hillslopes; thus, whole hillslopes may respond differently than the plots. Future rainfall simulation studies should be undertaken with large events at lower intensities to clarify the relationships between intensity, soil moisture storage and preferential flow and lateral flow generation.

At the Panola study hillslope in Georgia, where the K_s of the bedrock is 5.8 mm h^{-1} , lateral flow occurred when precipitation intensity was greater than 8 mm h^{-1} (Tromp-van Meerveld *et al.*, 2007). Bedrock infiltration accounted for 20–70% of the rainfall of large storms (Tromp-van Meerveld *et al.*, 2007). Similarly, Kienzler and Naef (2008) found that bedrock infiltration was greater for lower intensity applications and accounted for 30–50% of applied water. Lehmann *et al.* (2007) conducted modelling experiments on Tromp-van Meerveld *et al.*'s (2007) study site, which indicated there would be no lateral flow for events with a maximum precipitation intensity less than the K_s of the bedrock and that total hillslope outflow increased linearly with maximum precipitation intensity. On the hillslope in this study, lateral flow will not occur during precipitation events that are less than 10 mm h^{-1} in intensity. Even at these

intensities when soils are wet to field capacity, flow will be predominantly vertical.

This study further illustrates the interaction between soil storage and rainfall intensity in generating lateral flow from Boreal Plains hillslopes. During this experiment, soils were wet to field capacity prior to event 8—the event with the lowest applied precipitation intensity, and consequently, only small amounts of lateral flow (Redding and Devito, 2008). While this may be indicative of the importance of precipitation intensity, it is somewhat confounded by the fact that the rainfall depth of event 8 was approximately at the threshold necessary for lateral flow generation. The lack of observable lateral flow may also reflect the importance of roots or soil fractures in moving water vertically in the soil profile. This indicates that even at higher intensities, there may not have been enough surplus water beyond internal storage and rapid vertical flow to generate significant lateral flow.

The importance of the interaction of soil storage and rainfall intensity for exceeding lateral flow thresholds is further illustrated in field observations with natural rainfall. Prior to the irrigation experiment, a rainfall event of 27.5 mm with a maximum 30-min intensity of 5 mm h^{-1} fell on the RE plot soils, which were at field capacity, and failed to produce lateral flow (Redding and Devito, 2008). Furthermore, during July 2005, a rain event of approximately 80 mm over 23 h, with a maximum 30-min intensity of 24 mm h^{-1} , resulted in an increase in soil moisture storage of 65–80 mm at different monitoring locations (Figure 10) with no evidence of hillslope contributions to receiving wetlands or ponds. Although antecedent soil storage was low (Figure 10), the lack of observable lateral flow may also reflect the importance of roots or soil fractures in moving water vertically in the soil profile. Nevertheless, the overall event intensity of $>3 \text{ mm h}^{-1}$ for 23 h has

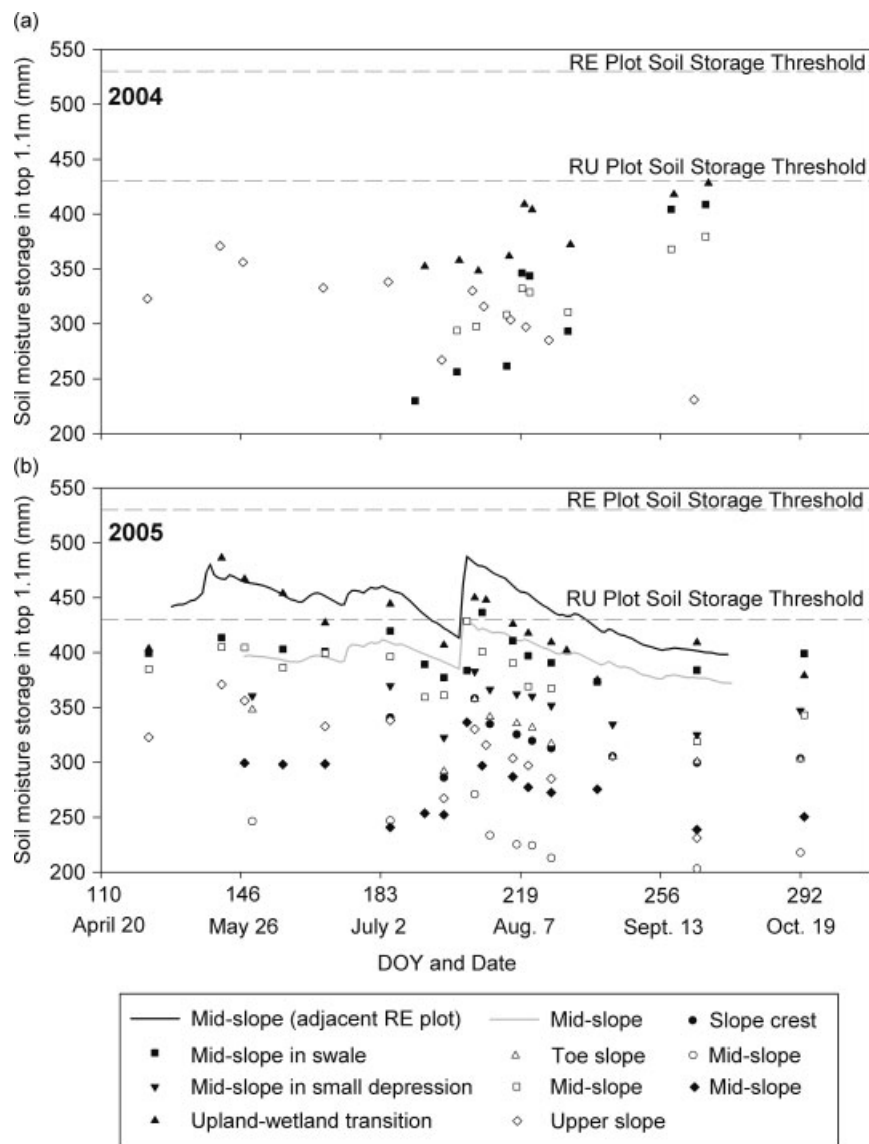


Figure 10. Measured soil moisture storage in top 1.1 m for the 2004 and 2005 growing seasons for the experimental hillslope and similar adjacent hillslopes. Individual symbols indicate sites with periodic manual TDR measurements. The black and grey lines are daily average soil moisture storage at sites with continuous measurement systems. Horizontal dashed lines indicate the storage thresholds required for lateral flow generation at the RE and RU plots based on the results of Redding and Devito (2008)

a return period of approximately 100 years (Hogg and Carr, 1985). This event, which was large for the research area, indicates the limited potential for lateral flow on these hillslopes with considerable soil storage capacity and known rainfall intensities (Figure 10). To better understand the role of precipitation intensity on lateral flow generation on these hillslopes, it will be necessary to investigate the response of precipitation applications well above threshold amounts at a range of intensities.

Spatial variability of soil properties

The influence of spatial heterogeneity on variability in runoff threshold and amounts is critical in modelling and predicting hillslope responses. Although plots in this study were only 4 m apart, and similar flow mechanisms were observed once the soils were wet to field capacity, the lateral flow responses to the later irrigation events ranged from runoff coefficients of 0.15 to 0.35 in the RU and RE plots, respectively, even though similar irrigation volumes and intensities were applied. Runoff responses in this study were low compared with those on sites on the Boreal Shield that had soils with limited soil storage and relatively impervious bedrock. On those sites, runoff coefficients were 0.35–0.75 or greater (Hill *et al.*, 1999; Buttle and McDonald, 2002).

Several factors could explain the difference in flow rate and magnitude between the two plots in this study during the later irrigation events. Once saturated, the differences in the surface slope of the plots (RE: surface slope = 0.15; RU: surface slope = 0.09) may have influenced the partitioning between vertical or lateral flow. In the RE plot, the higher clay content may have reduced the drainable porosity, thereby resulting in greater lateral flow due to reduced storage. After 2 weeks of post-irrigation drainage, changes in soil moisture storage in the top 1 m of the RU plot (–47 mm) were greater than those in the RE plot (–27 mm) (T. Redding, unpublished data). The longer period of wetting in the RE plot likely resulted in deeper wetting, thereby reducing the vertical water potential gradient that would have pulled water deeper. The RU plot had less wetting; therefore, the vertical water potential gradient would have more strongly influenced vertical flow and may have resulted in a reduced lateral flow response (Redding and Devito, 2008). It is also possible that the rooting distribution or orientation was slightly different between the plots; therefore, fewer roots in the RU plot connected to the pit face and moved water laterally. This is supported by the fact that only six preferential flow outlets were observed in the RU plot pit face, whereas nine were observed in the RE plot pit face. Because of differences in plot size, and similar total volumes and timing of water applications, rainfall depth and intensity were greater on the RE plot. There were no systematic differences between plots in the relationship between flow velocity and rainfall intensity (Figure 9); therefore, higher intensities in the RE plot could partially explain the greater lateral flow on that plot.

Soil heterogeneity may increase variability in lateral flow at depth and also influence the potential lateral

connectivity along hillslopes. This will further complicate predictions of timing and magnitude of lateral flow from entire hillslopes or watersheds that have heterogeneous soils and surficial geology. Groundwater studies conducted adjacent to the experimental plots showed the presence of large areas of dense clay, which may act as a restricting layer on hillslope segments that had increased lateral flow potential (K. Devito, unpublished data). However, lateral flow contributions to the hillslope base and adjacent wetlands or aquatic systems require that clay-rich areas be connected to each other or that they follow the slope. Given the heterogeneous stratigraphy of the hillslopes, with differences in depth to restricting layer (which controls available storage) and differences in particle size (which influences hydraulic properties), whole hillslope connectivity may be elusive because areas of deeper or more vertically conductive soils may act as barriers to spatially continuous lateral flow (McNamara *et al.*, 2005).

Potential for lateral flow under natural conditions

On forested hillslopes with Luvisolic soils on the Boreal Plains, lateral flow generation is a function of available soil moisture storage to a given depth, K_s at the base of that depth, which forms the “effective” restricting layer, and precipitation depth and intensity. The minimum storage that must be satisfied [assuming soils are at field capacity (S_{FC})] is determined by the precipitation intensity required to generate either preferential flow from the surface or lateral flow through percolation excess and transmissivity feedback. The conditions required for lateral flow generation can be summarized as follows (Figure 1):

1. Soil moisture storage must be filled to field capacity of the soil (S_{FC}) above the effective restricting layer.
2. Event P depth must be greater than internal storage (15–20 mm).
3. Event P intensity must be high enough for preferential flow initiation in the upper horizons, or greater than K_s at the depth of the effective restricting layer.

There is, however, a low probability of satisfying each of these conditions (Figures 10 and 11).

Given that potential evaporation rates are equal to or greater than precipitation, and most rainfall occurs during the growing season on the Boreal Plains, the probability of generating a moisture surplus sufficient to wet soils to field capacity is relatively low during the growing season (Devito *et al.*, 2005a,b). Plant water demand is lowest between snowmelt and leaf-out of the aspen canopy, and after senescence of the canopy and before the onset of winter. While soils are typically at their wettest immediately following snowmelt, the probability of rainfall events that satisfy criteria 2 and 3 above are very low (Figure 11). Precipitation data from Red Earth Creek indicates that in May, there is only a 1.5% probability that daily precipitation will be 20 mm

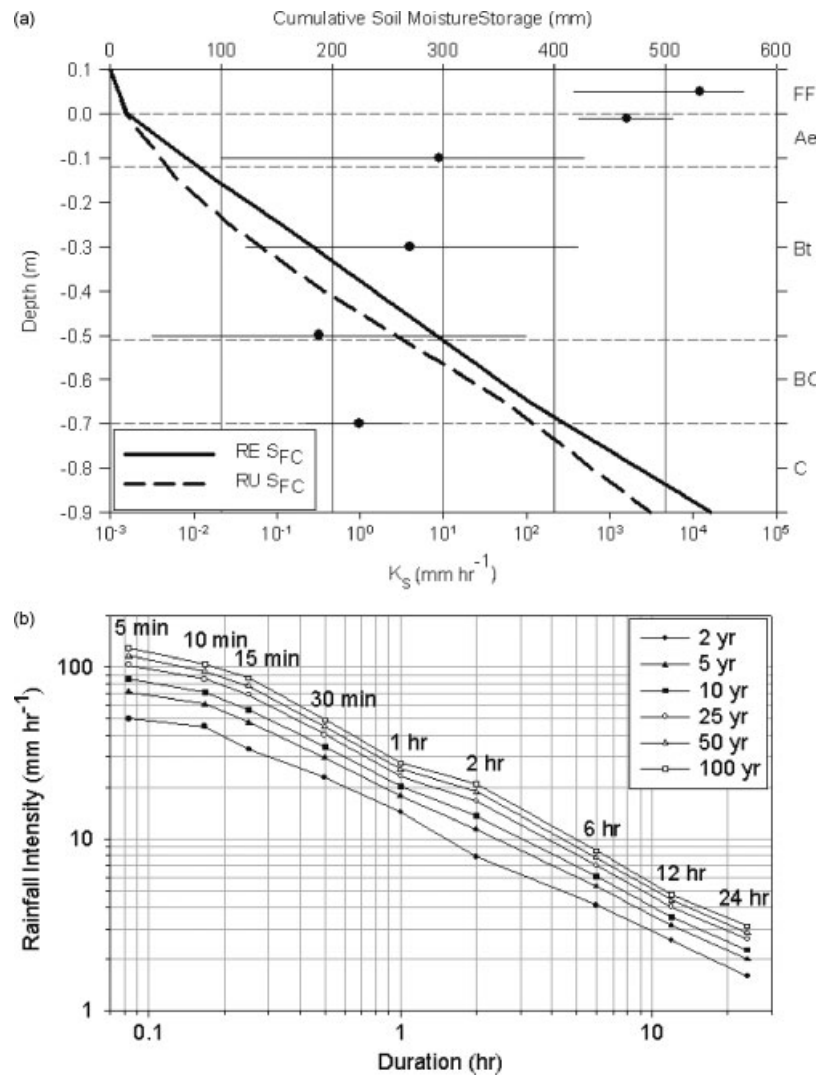


Figure 11. Plot showing cumulative available soil moisture storage at field capacity (S_{FC}) with depth for the root exclusion (RE) and root uptake (RU) plots. Estimates of S_{FC} are calculated from TDR data. Horizontal solid lines and filled circles are the range and geometric mean of measured saturated hydraulic conductivity (K_s) for given depth intervals. Plot (b) shows rainfall intensity–duration–frequency for the Utikuma Region Study Area region, taken from Hogg and Carr (1985). Each line is a separate return period (year)

or greater. In the fall, once the forest has senesced, soils are typically drier than they are post-snowmelt (Figure 10), and there is only a 1.1% probability that daily precipitation will be 20 mm or greater. Therefore, even if soils are at field capacity, which is uncommon, there is a low probability that daily precipitation will exceed the threshold required (Redding and Devito, 2008). It is also clear from the data presented in Figure 10 that many locations do not reach the soil moisture storage thresholds over the spring, fall or growing season. There is also considerable spatial variability between locations on the same or adjacent hillslopes with similar soils. During the 2005 growing season, some locations approached or exceeded the RU plot threshold but only following snowmelt and after a rain event of approximately 80 mm (Figure 10).

To evaluate the threshold-based conceptual model, depth profiles of available soil moisture storage and K_s , and precipitation intensity–duration–frequency curves for the study region are required (Figure 11). On the

basis of the model, rainfall generated lateral flow will occur only infrequently on moraine soils characteristic of the study region, along a continuum of precipitation intensity and soil moisture storage. The extremes of the continuum range from very high-intensity events that only need to saturate soils near the surface (shallow effective restricting layer) to lower intensity events on already wet soils (deeper effective restricting layer). In either case, the probability of exceeding the limiting factors (very high-intensity precipitation or filling soil storage to C horizon) is low. For example, to generate lateral flow from the Ae horizon, it is necessary to wet the profile to the C horizon and then have a precipitation event that provides adequate water to saturate the profile upwards from the restricting layer (C horizon) into the Ae horizon. The RU plot required 97 mm of applied precipitation before significant (>1 mm) lateral flow was generated. This was additional to considerable wetting that had occurred between the end of the growing season and the beginning of the irrigation applications (Redding

and Devito, 2008). Once soils are wetted to this depth, a rainfall event of 20 mm or greater is required to satisfy internal storage (Redding and Devito, 2008), and the intensity must be greater than the K_s of the C horizon (0.4–7 mm h⁻¹). The minimum return period observed is 2–5 years, which is for an event of 10 mm h⁻¹ for 2 h (Figure 11).

Alternatively, lateral flow may be generated from the Ae horizon if the required storage is filled (which is much less than filling storage to C horizon); however, the P intensity required to generate lateral flow by percolation excess must exceed the K_s of 10–400 mm h⁻¹. During the experiment, precipitation intensity of greater than 80 mm h⁻¹ was required to generate flow from the Ae horizon. The extreme rainfall intensities required under this scenario are highly unlikely. This also indicates that under unfrozen conditions, overland flow is not likely for these hillslopes.

Both runoff generation extremes outlined above will be infrequent based on the precipitation depths required to fill storage to the C horizon or on the high precipitation intensities required to exceed K_s near the surface. Because K_s decreases with depth, both lateral and vertical flow will be limited, and water will be stored and/or develop transient saturation. If transient saturation occurs, it will have to rise higher in the profile (transmissivity feedback) for lateral flow to occur through macropores. For most storms observed in the study region, the precipitation intensity is less than the K_s , and there is large available soil moisture storage in the unsaturated zone. Regional stream runoff responses indicate there is very little runoff (runoff coefficient less than 10%) in most years (Redding and Devito, 2008). Significant lateral flow is produced only infrequently, usually by large and intense rainstorms that follow wet years, such as those experienced in 1997 (Devito *et al.*, 2005a).

SUMMARY AND CONCLUSIONS

To predict when hillslope runoff and recharge may occur on the Boreal Plains in response to rainfall events, it is necessary to understand the thresholds, mechanisms and flowpaths of lateral and vertical flow. Limited sampling under natural conditions has not allowed for adequate testing of hypothesized flowpaths and mechanisms for Luvisolic soils on the Boreal Plains. In contrast, this study highlights the power of manipulative experiments in directly testing flowpaths and mechanisms. Manipulative experiments are especially important for understanding the thresholds of hydrologic response in environments where they occur infrequently under natural conditions (Kirchner, 2006).

Two complementary mechanisms produced lateral flow: the first was preferential flow generated from near the soil surface; the second involved the development of a transient perched water table above the C horizon and the generation of lateral flow by the transmissivity feedback mechanism. At the precipitation event scales tested,

lateral subsurface flow paths were dominated by preferential flow through root channels in the Bt horizon rather than by flow at the Ae–Bt horizon boundary.

The controls on lateral flow generation are antecedent soil moisture storage, soil K_s , event P depth, and event P intensity. For areas without a well-defined restricting layer (e.g. impervious bedrock), which directs flow laterally, the layer upon which lateral flow develops is defined by the hydraulic conductivity of that layer relative to the rate of water input. In this study, hydraulic conductivity below approximately 0.5 m depth declined to a rate of <10 mm h⁻¹. Given the high-intensity of applied rainfall, lateral flow occurred above this layer, through the Bt horizon.

On the basis of the climate and soil characteristics of the study site, there is a low potential for lateral flow from similar hillslopes in the Boreal Plains region. In wet years, the potential for lateral flow increases but remains low due to the high ET demand of the forest vegetation during the summer rainy season. To understand and predict the frequency of lateral flow and drainage from the soil profile, it will be necessary to combine an understanding of frequency distributions of precipitation and soil moisture storage. Long-term data on soil moisture conditions are not available for this area; therefore, models will have to be used to generate the probability distributions of soil moisture conditions. Future research should examine the potential for lateral subsurface flow during snowmelt on these hillslopes.

On the basis of the rainfall return periods required to generate lateral flow from the study plots, it appears that forested hillslopes with fine-textured soils rarely contribute lateral flow and are, therefore, decoupled from adjacent wetlands and ponds most of the time. This has important implications for the hydrology and biogeochemistry of wetland systems in this region, and for the sustainability of wetland habitats under conditions of climate change or watershed disturbance.

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